CSE 4345: Big Data Analytics

Week 2, Part 1 — Challenges of Big Data: Extraction, Integration & Heterogeneous Information

Department of Computer Science & Engineering
Premier University, Chittagong

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Course & Lecture Information

Lecture Identity

Course:	CSE 4345
Week / Part:	2 of 11 | Part 1 of 2
CLO:	CLO2
PLO:	PLO(b), PLO(c)

CLO2

"Analyse the existing challenges in Big Data"

Prerequisite Recall

Week 1 established the **3Vs**—Volume, Velocity, Variety. Week 2 dives into **Variety** in depth: what happens when you must integrate data from tens of heterogeneous sources.

Part 1 Covers

- The data silo problem
- Four types of heterogeneity
- ETL: Extract, Transform, Load (in depth)
- ELT: The data lake paradigm
- Data quality dimensions & provenance
- Entity resolution
- Ivy League alignment
- Student assessment pool

Part 2 Preview

Seminal paper: Halevy, Rajaraman & Ordille (2006)
Case Study: HL7/FHIR & Healthcare Interoperability

Lecture Agenda

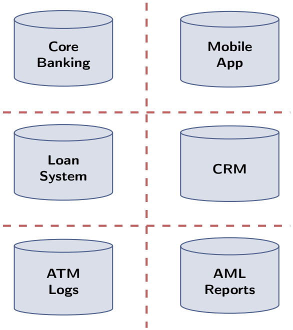
The Data Silo Problem

Organisations Are Data-Rich, Insight-Poor

The Reality of Enterprise Data (2024)

A mid-sized bank in Bangladesh typically holds customer data in:

- **Core banking system** (Oracle RDBMS)
- **Mobile banking app** (MySQL + NoSQL)
- **Loan origination system** (separate legacy RDBMS)
- **Call centre CRM** (Salesforce cloud)
- **ATM transaction logs** (flat CSV files, nightly dump)
- **AML / fraud reports** (Excel spreadsheets)
- **Regulatory filings** (Bangladesh Bank XML schemas)



Data Silos — No Unified View

A 2019 Forrester Study

73% of enterprise data is never analysed. The primary reason: **integration complexity**.

The Integration Challenge at Scale

The Core Problem Statement

"Given a set of heterogeneous, autonomous data sources, provide the user with a ***unified, transparent query interface*** *that gives a single coherent view of all data."*

— Halevy, Rajaraman & Ordille (2006)

Three fundamental dimensions of difficulty:

1. **Heterogeneity**: Sources use different schemas, formats, vocabularies, and update cadences
2. **Autonomy**: Each source is independently managed; the integration layer cannot change the source
3. **Scale**: The volume and velocity of data make manual reconciliation impossible

Challenge	**Big Data Dimension**
Heterogeneity	Variety — different formats, schemas
Scale	Volume — 100 TB cannot be loaded into memory
Noise & gaps	Veracity — sensor dropout, OCR errors
Provenance	Regulatory — GDPR, PDPA
Latency	Velocity — real-time vs. nightly batch

Key Insight

The 3Vs from Week 1 were symptoms. The

Types of Data Heterogeneity

Four Dimensions of Heterogeneity

Structural / Schema
Different column names
& table organisation

Semantic
Same name,
different meaning

Heterogeneity

Syntactic / Format
Same data,
different encoding

Scale / Granularity
Different resolutions,
units, cardinalities

Why All Four Matter Simultaneously

A single integration pipeline must handle all four dimensions at once. Solving only structural heterogeneity (schema matching) while ignoring semantic heterogeneity (meaning) produces a technically unified but analytically **misleading** dataset.

Structural (Schema) Heterogeneity

Definition

Two data sources represent the **same real-world concept** using **different table structures, column names, or normalisation levels**.

Two sub-types:

- **Naming conflicts**: Different names for the same attribute
  - Source A: `cust_id` ↔ Source B: `customer_number`
  - Source A: `dob` ↔ Source B: `birth_date`
- **Structural conflicts**: Flat vs. normalised schemas
  - Source A: one `address` column (full address as a string)
  - Source B: four columns (`street`, `city`,

Bangladesh NID Integration Example

	BREC Schema	**MOHFW Schema**
ID field	`nid_no`	`national_id`
Full name	`full_name`	`fname + lname`
Birth date	`dob`	`birth_dt`
Address	Single string	5 separate columns
Gender	`M/F`	`1/2`

Integrating Bangladesh Election Commission records with Ministry of Health records requires **mapping every field pair** before any query can span both.

Semantic Heterogeneity

Definition

Two data sources use the **same attribute name** (or structure) but **mean different things**, or use different vocabularies for the same concept.

Common manifestations:

- **Homonyms**: Same word, different meaning
  - `date` = admission date in Hospital A
  - `date` = birth date in Hospital B
  - `date` = record-entry date in Health Ministry
- **Synonyms**: Different words, same concept
  - `revenue` vs. `income` vs. `turnover` (same KPI, different label)
- **Granularity mismatch**: "Region" = division in one system, district in another
- **Unit conflicts**: Temperature in °C vs. °F;

Why Semantic Heterogeneity is Hardest

Structural heterogeneity can often be resolved **algorithmically** (match column names, infer types). Semantic heterogeneity requires **domain knowledge**—a machine cannot know that `date` means birth date without reading the data dictionary.

Real Impact: Challenger Disaster (1986)

NASA and Morton Thiokol engineers used the word "acceptable" to mean different risk levels when discussing O-ring failures. This semantic mismatch contributed to the decision to launch.

Source: Vaughan, D. (1996). The Challenger Launch Decision. U. of Chicago Press.

Syntactic & Scale Heterogeneity

Syntactic / Format Heterogeneity

The **same data value** is encoded in **different formats** across sources.

Examples (all representing January 15, 2024):

Format	Value
ISO 8601	2024-01-15
US format	01/15/2024
Bangladesh	15-01-2024
Unix epoch	1705276800
Bengali calendar	01 Magh 1430

Beyond Dates

Phone numbers: 01711 123456 vs

Scale / Granularity Heterogeneity

Sources represent the **same domain** at **different levels of detail** or resolution.

Examples:

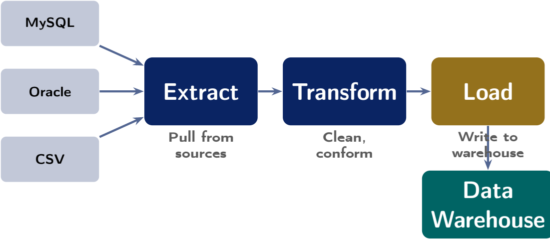
- **Geographic**: One source records GPS coordinates (lat/long to 6 decimal places); another records only the district name
- **Temporal**: Sales DB records hourly totals; ERP records daily totals; financial system records monthly summaries
- **Numeric precision**: Sensor A: temperature to 0.001°C; Sensor B: rounds to nearest integer
- **Aggregation**: One system reports

ETL and ELT Paradigms

The ETL Process: Origin and Logic

What is ETL?

Extract – Transform – Load: the classical approach to data integration, dominant from the 1990s through the 2010s. Data is **transformed before it reaches the warehouse**.



Why ETL was designed this way:

- Storage was **expensive**—only clean, structured data deserved to be stored
- Data warehouses (Teradata, Netezza) had **rigid schemas** that required pre-conformed data
- Analysts knew *upfront* exactly which reports they needed—the schema was designed for known queries
- Regulatory compliance demanded **verified,**

Key Characteristic

The data warehouse only ever sees **clean, structured, schema-conformed** data. Raw data never enters.

ETL: The Three Phases in Depth

Phase 1: Extract

Pull data from source systems:

- **Full extract**: dump entire source table (expensive, nightly)
- **Incremental extract**: only changed rows since last run (CDC — Change Data Capture)
- **Connectors**: JDBC for RDBMS, REST APIs, SFTP for file drops
- Challenge: source systems may be **unavailable** during extraction window

Phase 2: Transform

The most **complex** stage:

- **Cleansing**: fix nulls, remove duplicates, correct typos
- **Standardisation**: unify date formats, currency, units
- **Schema mapping**: rename columns, restructure tables
- **Enrichment**: join with reference tables (e.g., add district name from postcode)
- **Aggregation**: compute summaries (daily totals from raw transactions)
- **Validation**: reject rows that violate business rules

Phase 3: Load

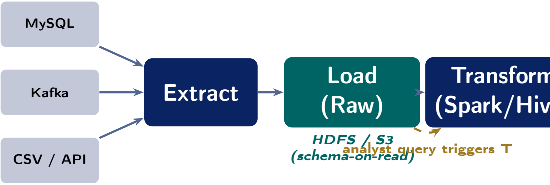
Write transformed data to the target:

- **Full load**: overwrite the target table (simple, but slow)
- **Incremental load**: append or upsert new/changed rows
- **Slowly Changing Dimensions (SCD)**: preserve historical versions of records (e.g., a customer's old address)
- **Partitioned load**: write to date/region partitions for query performance

ELT: The Data Lake Approach

What is ELT?

Extract – Load – Transform: Load raw data *first* into a cheap, schema-flexible storage layer, then transform *on demand* using Spark or Hive.



Why ELT emerged (post-2010):

- Cloud storage costs dropped to **$0.023/GB/month** (AWS S3)—cheap enough to store raw data
- **Schema-on-read** (Hive, Spark): define schema at query time, not ingest time
- Data scientists need **raw, unprocessed** data for ML feature engineering—ETL discards too much
- Transformation requirements are often **unknown** at ingest time (exploratory analytics)

Key Characteristic

Raw data **always preserved**. Transformation is triggered *by the query*, not at ingest time. Multiple analysts can apply different transformations to the same raw data.

ETL vs. ELT: Comprehensive Comparison

Dimension	**ETL**	**ELT**
Transform timing	Before loading (at ingest)	After loading (on query)
Schema	Schema-on-write (rigid)	Schema-on-read (flexible)
Storage	Only clean data stored	Raw *and* processed data stored
Raw data	Discarded after transform	Always preserved in data lake
Latency	High (transform pipeline runs before data is queryable)	Low (data available immediately after load)
Best for	Known query patterns, compliance, sensitive data	ML feature engineering, exploration, unknown future queries
Tools	Informatica, SSIS, Talend, Ab Initio	Apache Spark + HDFS/S3, AWS Glue, dbt
Example	Nightly bank reconciliation to	Netflix data lake: raw events

Data Quality, Provenance & Entity Resolution

Data Quality: Six Dimensions (DAMA Framework)

Dimension	Definition	Big Data Example
Accuracy	Correctly reflects reality	GPS coordinates drifting 200 m
Completeness	All required data present	30% of sensor readings missing due to network dropout
Consistency	Data agrees across systems	Customer age is 32 in CRM but 35 in loan system
Timeliness	Data is current for use case	Yesterday's stock price used for real-time trade decision

The "Garbage In, Garbage Out" Law

A sophisticated machine learning model trained on inaccurate, incomplete, or inconsistent data will produce **confidently wrong** predictions. Data quality validation must happen **before** any analytics.

Cost of Poor Quality

IBM (2016) estimated that poor data quality costs the US economy **$3.1 trillion per year** in lost productivity, regulatory penalties, and incorrect decisions.

Source: IBM Big Data Hub, 2016

Provenance: Knowing Where Your Data Came From

Definition of Data Provenance

Data provenance (also called *data lineage*) tracks the **origin, transformation history, and movement** of a data item from its source to its current form.

Two types of provenance:

- **Where-provenance**: Which source record(s) produced this value?
  - e.g., This patient's age came from the NID database on 2024-01-10
- **How-provenance**: What transformations were applied?
  - e.g., Salary field = `gross_pay` − `tax` − `pf_deduction` (per Spark job v2.3.1, run 2024-01-15)

GDPR Article 30: Mandatory Provenance

The EU General Data Protection Regulation requires organisations to maintain **Records of Processing Activities**—effectively a provenance log:

- Where did this personal data come from?
- Who has accessed it?
- What transformations were applied?
- Where was it sent (third parties)?
- When will it be deleted?

Penalty for non-compliance: Up to **4% of global annual revenue** or €20 million (whichever is higher).

Entity Resolution: Linking Records Across Sources

Definition

Entity resolution (also: record linkage, deduplication, identity resolution) determines whether two records from **different sources** or **within one source** refer to the **same real-world entity**.

Why it is hard:

- Name variations: "Mohammad Ali" vs. "Md. Ali" vs. "M. Ali"
- Typos: "Chittagong" vs. "Chattogram" vs. "Chittagram"
- Missing fields: no NID number in one source
- Identical names: 500 different "Rahim Uddin" records in a district

Four-Step Entity Resolution Pipeline

1. Blocking
Group by postcode / name prefix

↓

2. Feature Extraction
Edit distance, Soundex, Jaccard

↓

3. Scoring
Similarity threshold or ML model

↓

4. Resolution
Merge / link / flag for review

Blocking reduces complexity

Comparing only within blocks of records sharing the same first-3 letters of surname reduces $O(n^2)$ to $O(n \log n)$ in practice.

Ivy League Alignment

Data Integration in Top University Curricula

MIT 6.830 Database Systems (Madden, Stonebraker)

MIT's graduate database course frames data integration as one of the **three unsolved grand challenges** in data management (alongside query optimisation and transaction management at scale).

- **LAV vs. GAV**: Two competing approaches to schema mapping
  - **GAV (Global-as-View)**: Define global schema, write queries over sources
  - **LAV (Local-as-View)**: Define each source as a view over the global schema
- **Pay-as-you-go integration** (Halevy, 2006): Integrate incrementally as queries arrive, rather than resolving all heterogeneity upfront

Stanford CS246 Perspective

Leskovec et al. open their course with: *"Before you can mine data, you must **collect** it, **clean** it, and **integrate** it. In practice, 80% of a data scientist's time is spent on these steps."* **The 80/20 Rule of Data Science:**

- **80%** of effort: data collection, cleaning, integration
- **20%** of effort: modelling, analysis, visualisation

This is why CLO2 (*analyse integration challenges*) is foundational—you cannot reach CLO3/CLO4 without solving it.

Student Assessment Pool

Assessment—MCQ Set A

Instructions

Select the single best answer. Correct answers **bolded** for instructor reference.

1. Which data integration technique determines whether two records from **different datasets** refer to the same real-world entity?
   a. Schema mapping (b) Data exchange (c) **Entity resolution** (d) Data partitioning

2. In the **ELT** paradigm, when does the transformation step occur?
   a. Before data extraction (b) During the extraction process (c) Before loading into storage (d) **After data is loaded, on demand at query time**

Instructor Notes

Q1: "Data exchange" (option b) is a common confusion—it means materialising data from source to target schema, not linking records. **Q2:** Options (a) and (c) describe ETL; students must clearly distinguish ELT's defining inversion.

Assessment—MCQ Set B

CLO2

Q3. Which data quality dimension checks that each real-world entity is represented **exactly once** in a dataset?

(a) Accuracy (b) Completeness (c) Consistency (d) **Uniqueness**

Q4. A healthcare database uses `date` to mean *admission date*, while a pharmacy database uses `date` to mean *prescription date*. What type of heterogeneity is this?

(a) Structural heterogeneity (b) Syntactic heterogeneity (c) **Semantic heterogeneity** (d) Scale heterogeneity

Instructor Notes

Q3: Students often confuse uniqueness with consistency. Consistency = data agrees *across* systems; uniqueness = no duplicates *within or across* systems. **Q4:** The field name (`date`) is the same in both—this is a classic homonym/semantic conflict. Structural heterogeneity would be different column *names* for the same concept.

Instructions

State True or False and provide a **one-sentence justification** for full marks.

Statement	**Answer**
1. ETL is the preferred paradigm for modern data lake architectures because it enforces schema early.	**False**
2. Semantic heterogeneity occurs when the same attribute name carries different meanings in different source systems.	**True**
3. In ELT, the transformation step is applied *before* data is written to the storage layer.	**False**
4. GDPR Article 30 effectively mandates that organisations maintain data provenance records for personal data.	**True**

Assessment—Descriptive Questions *CLO2*

Instructions

Answer each question in **100–150 words**. Use specific terminology from the lecture.

- D1. Define the **ETL process** and explain each of its three phases (Extract, Transform, Load) with a concrete Big Data example from the banking or healthcare sector in Bangladesh.
- D2. What is **schema heterogeneity**? Distinguish clearly between *structural* heterogeneity and *semantic* heterogeneity, and provide one specific example of each drawn from a multi-hospital patient data integration scenario.
- D3. Identify and describe **three data quality dimensions** from the DAMA framework. For each, describe a concrete failure scenario in a Big Data pipeline and a mitigation strategy.

Assessment—Analytical Questions *CLO2*

Instructions

Answer in **200–300 words**. Show step-by-step reasoning and reference specific tools or concepts from the lecture.

A1. **Integration Architecture Design:**
A Bangladeshi government agency wants to build a unified public health analytics platform by integrating data from 64 district hospitals, each using a different EHR system (some MySQL, some paper-based digitised as CSV, some using a custom XML API).
  - Identify **four specific integration challenges** from today's lecture that this agency will face.
  - For each challenge, propose a **concrete technical strategy** to address it.
  - Should this agency use **ETL or ELT**? Justify with reference to their data characteristics.

A2. **Entity Resolution at Scale:**
A national mobile financial services provider (like bKash) wants to merge its user database (50 million records) with Bangladesh Bank's credit history registry (40 million records). Explain why a naïve pairwise comparison is computationally infeasible, then design a **blocking strategy** and **similarity scoring approach** to perform this entity resolution at scale. What data quality dimension(s) would you validate *after* resolution is complete?

Textbook & Reading References

Primary Textbooks for Week 2

- **[SA]** Acharya & Chellappan. *Big Data and Analytics*, Wiley, 2015. **Chapter 2**
- **[TE]** Erl, Khattak & Buhler. *Big Data Fundamentals*, Prentice Hall, 2016. **Chapter 3**
- **[LRU]** Leskovec, Rajaraman & Ullman. *Mining of Massive Datasets*, 3rd ed., 2020. **Chapter 1.3**

Supplementary (Covered in Week 2, Part 2 — Seminal Papers)

- Halevy, A., Rajaraman, A., & Ordille, J. (2006). **Data Integration: The Teenage Years.** *Proceedings of VLDB*. **[PRIMARY SEMINAL PAPER]**
- Doan, A., Halevy, A., & Ives, Z. (2012). **Principles of Data Integration.** Morgan Kaufmann. *[Graduate textbook at MIT, Stanford]*
- Kleppmann, M. (2017). **Designing Data-Intensive Applications.** O'Reilly. Ch. 10–11. *[Highly recommended supplementary reading]*

For DAMA Data Quality Framework

DAMA International (2017). *DAMA-DMBOK: Data Management Body of Knowledge*, 2nd ed. Ch. 13 (Data

Questions & Discussion

Week 2, Part 1 | CSE 4345 Big Data Analytics

"Data is the new oil. But crude oil must be refined before it can power anything. Data integration is that refinery."

—Adapted from Clive Humby (2006)

Next Lecture: Week 2, Part 2

Seminal Paper: Halevy, Rajaraman & Ordille (2006) — *"Data Integration: The Teenage Years"*

Schema Mapping · LAV vs. GAV · Pay-As-You-Go Integration
Case Study: HL7/FHIR & Healthcare Interoperability ($8.3B annual cost of failure)

Reading before Part 2: [SA] Ch. 2 · **[TE]** Ch. 3